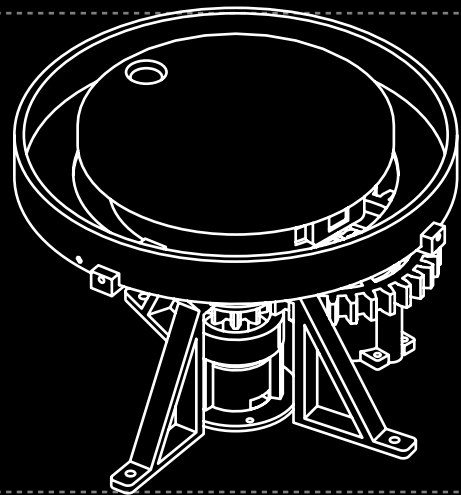
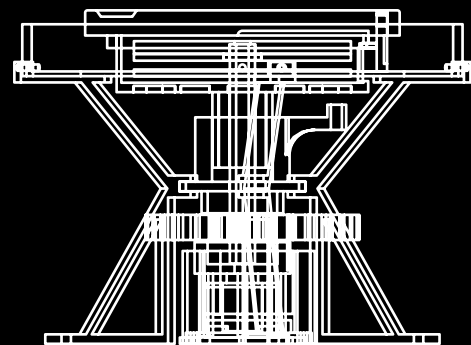


GIZMOPHONE

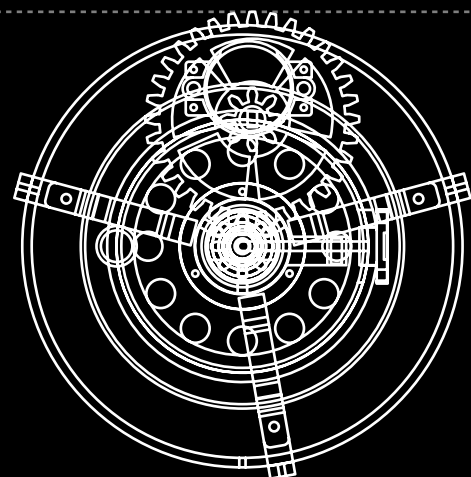
‘Rhythm as Rotation’



The gizmophone is an instrument designed to challenge traditional musical notation by using a rotating face as an interface to display rhythm and melody.



It creates a more fluid musical experience which relies on visual, and spacial awareness, as well as pattern recognition rather than the traditional arbitrary leading lines or bars.



The instrument is composed of 3 parts:
Rhythm as rotation
Colour as pitch/timbre
Tempo as a hand of a clock



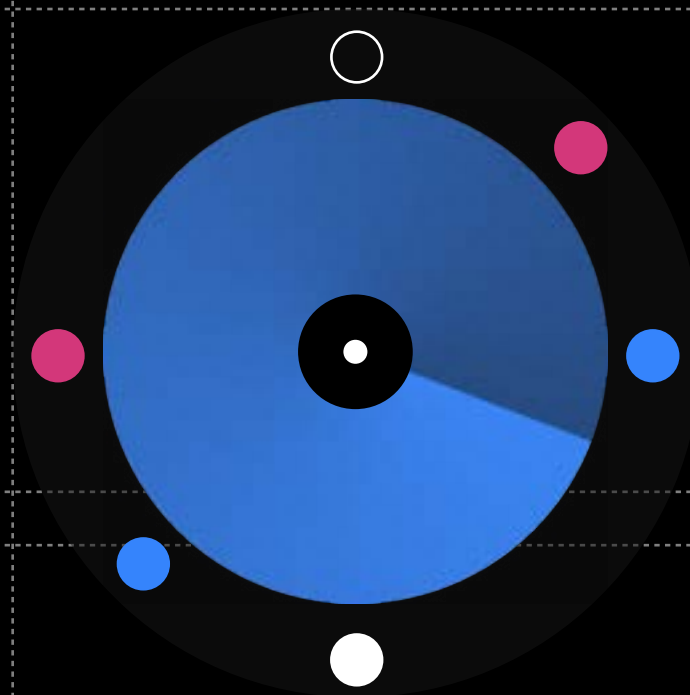
GIZMOPHONE

Interaction Design and Motivation



Rhythm as a Clock

A regular 'clock tick' creates an intuitive metronome, where the summative removal or replacement of these 'ticks' can be used to create complex rhythmic patterns. This replaces abstract, linear musical bars with a rotation around a circle (a much more intuitive, visual experience which relies less on musical theory knowledge). It fosters a more fluid musical approach relies on visual, and spatial intuition, and pattern recognition to create a rhythm.



Creating Melody

By replacing our 'ticks' with coloured pegs, we can now visualise pitch and timbre, transforming a rhythmic clock into a melodic instrument! In our project, we will assign separate percussive instruments to each colour to demonstrate this idea.



Speed

We can include speed by creating a manual interaction with the hand of our clock. When we spin the hand, we determine how many rotations per minute we would like, and therefore the speed of the melody we play. For rests, we can use gaps between pegs, or hold the hand of our clock. If we need to change the tempo, we manually grab the hand to reset it.

- Dominant
- Tonic
- Mediant
- Subdominant
- Subtonic
- Octave 1
- Octave 2
- Octave 3

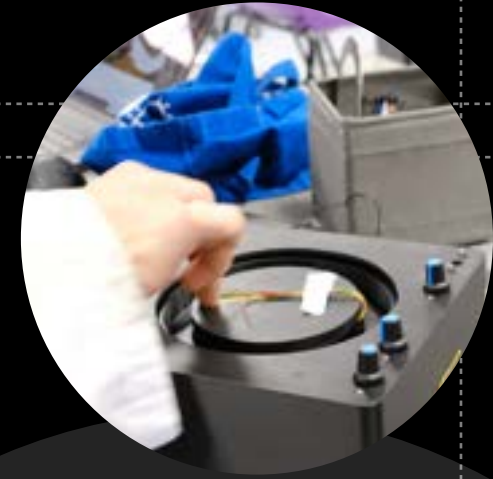
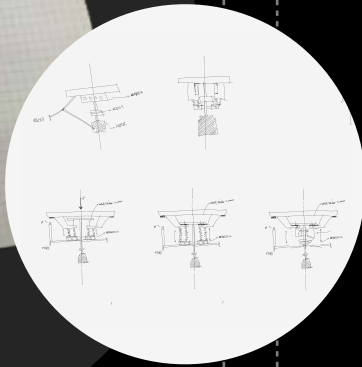
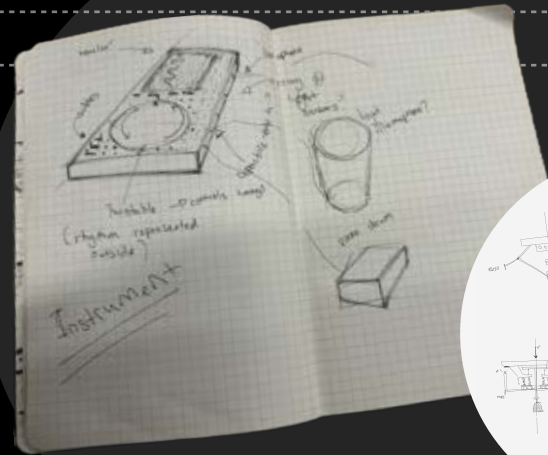


Developing Further

Including more musical theory augments the system further. New rings for instruments, representing notes in the scale as colours on the pegs, then colour of the border as the octave we are playing. Now make the pegs smudges and you have sustained notes!

GIZMOPHONE

Design Process



Initial Concept

The first idea of the Gizmophone was quite different from the final outcome. Although it shares the same turntable interaction, it did not include 2 primary components of the idea: colour-based beat triggering and the clutch mechanism that allows manipulation of rpm.

Version 1

This was the initial attempt, it allowed us to prototype mechanics, trial bearing types and test materials that could be used for the clutch pads.

Version 2

The next major design overhaul was to integrate the motor drive system. Version 2 served as an important and adaptable testbed for (a lot of) gear ratios, types and arrangements.

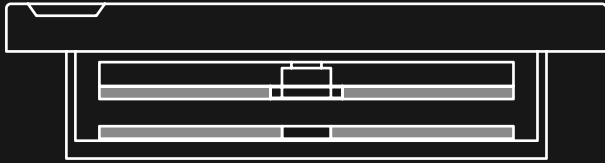
It also let us add to the mechanical design, with the need for a slip ring, top 'interaction disk' and a mounting system for the coloured pegs.

Version 3

The final version of the Gizmophone includes double helical gears, sturdier legs, and a full enclosure. The housing primarily serves as mounting for electronic components and circuitry.

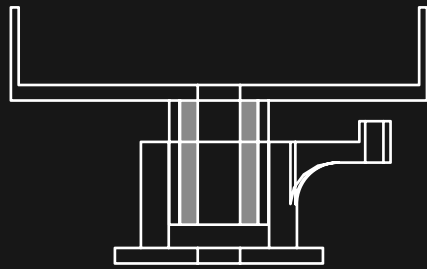
GIZMOPHONE

Mechanical Design



Clutch

The clutch mechanism is used to decouple the motor from the interaction disc (our clock hand), which is necessary for our start up sequence. It is composed of a drive shaft, an upper clutch plate, and a lower clutch plate which attaches to the interaction disk.



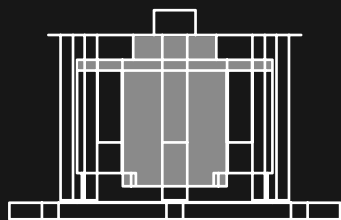
Bearing + Spring

The bearing assembly attaches to the lower clutch plate, and slots inside a spring holder. Attached to the underside of the plate is an array of magnets, and attached to the spring holder is a hall effect sensor. This array allows us to measure the rotation of the disk, and also depress the interaction disk to decouple it.



Gears + Gear Ratio

To up the speed of our stepper motor we created a 4:1 double helical gear chain. This set of gears proved most efficient for its self levelling capabilities. We also tried driving with a dc motor, however this was significantly louder, and stepping up the rpm was more efficient than stepping down with 3d printed gears.



Slip Ring Assembly

Lastly, we connected a slip ring to the bottom of the shaft to allow the wires connected to the colour sensor to spin with the disk. On top of the split ring assembly is a bearing holder, and shaft stopper, which stops the shaft from depressing onto the slip ring and allows it to spin more freely at its base.



Coloured Pegs

We 3D printed coloured pegs with integrated internal magnets. Under the outer disc and on the side of the final project, we used mild steel to magnetise the surface. This was used to give the 'primary tool' of our interaction a satisfying snap. It was important that we had a continuous magnetic surface on the bottom of the disc, as it gives more musical freedom to the user, allowing any time signature or to syncopate and swing rhythms.

GIZMOPHONE

Electronic Hardware and Circuitry

TCS34725

Colour Sensor

Justification

A very small package that contains strong LEDs, this is great for our application as the sensor needs to be spun. Our sensor also has a stacked board with solderable connections on the opposite side, further minimising it's size.

A3144

Hall Effect Sensor

Justification

Reliable and easy to source sensor with good physical range, to detect rpm while the clutch is either engaged or disengaged. This means we don't have to physically move the sensor.

28BYJ-48

Stepper Motor

Justification

We used a stepper over DC motor for consistency in 'bar' length, this model also has the ability to use a half step sequence for smoothness.

DMA45-4 1-1/2"

Speaker Driver

Justification

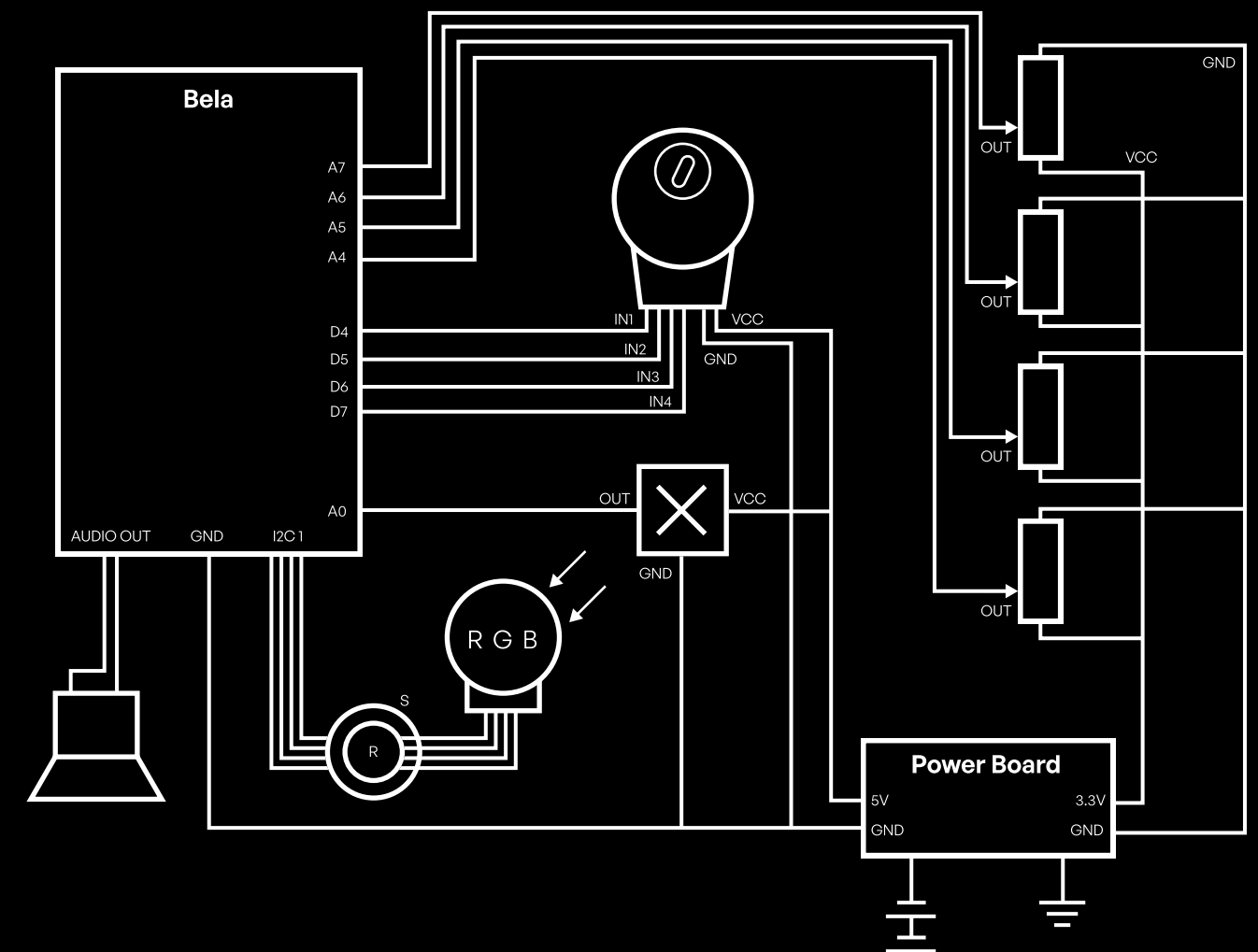
A small driver that we could easily fit, while working seamlessly with Bela's amps and small power output.



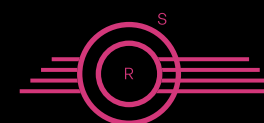
Potentiometers - Volume

To control the volume of each individual drum, we used potentiometers connected to analog pins, detecting the divided voltage and mapping it to the volume of the sound through PureData Variables.

Circuit Diagram



Additional Hardware



Slip Ring

This allows us to mount the colour sensor on the spinning elements, it carries 4 wires, for the I2C connection.



Magnet Array

The array of magnets consists of 12 evenly spaced, giving us a time/speed measurement for every 30° rotated.